

EVALUATION OF THE IMPACT OF A CARDIOVASCULAR DISEASE PREVENTION PROGRAM ON COMMUNITY HEALTH



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01. INTRODUCTION

Cardiovascular disease (CVD) remains a leading cause of morbidity and mortality worldwide. According to the World Health Organization, noncommunicable diseases (NCDs) now account for the vast majority of global deaths, of which CVDs are the principal contributors. The WHO emphasizes that many CVDs are preventable by addressing modifiable risk factors, such as hypertension, obesity, unhealthy diets, physical inactivity, smoking, and uncontrolled diabetes.

Globally, aging populations, urbanization, and changing lifestyles have led to escalating burdens of cardiovascular risk. The WHO states that targeted community-based strategies such as health education, regular screening, lifestyle counselling, and promotion of physical activity are pivotal in changing the trajectory of CVDs.

In Sri Lanka, over recent decades, the proportions of deaths due to cardiovascular conditions has increased from about 24.9% in 2004 to over 28.1% by 2019. The country also faces the burdens of various risk factors such as hypertension, obesity, diabetes, and tobacco use whose prevalence has been rising, particularly in urban populations.

Since heart disease is escalating into such a big problem, community-based prevention programs are more important than ever. Around the world, programs that include health education, regular check-ups, lifestyle advice, and group exercises have been shown to lower the chances of serious heart problems. Bringing similar programs into Sri Lanka's districts can play a key role in reducing future heart disease and protecting the health of local communities.

The present study is an 18-month community-based CVD prevention program done by the Ministry of Health in Sri Lanka launched across three districts Colombo, Kandy, and Galle. The program brought together several activities all aimed at helping people lower their risk of heart disease, adopt healthier habits, and improve their overall heart health.

This study uses the before-and-after evaluation data to assess changes in cardiovascular risk factors such as systolic blood pressure, body mass index, smoking, and physical activity and to quantify the incidence of subsequent CVD events. In doing so, it also examines differential effects by demographic subgroups such as district, age, gender, education. The findings are intended to provide scale-up strategies for community-based cardiovascular prevention in Sri Lanka and similar low- and middle-income settings.

02. Objectives of the study

Evaluate Program Impact on Risk Factors and Outcomes

- Assess changes in systolic blood pressure, BMI, and smoking status before and after the program.
- Measure the effect of the program on exercise frequency and diabetes prevalence.
- Estimate the incidence of cardiovascular (CVD) events during the follow-up period.

Compare Subgroup Differences

- Analyze whether program impact differed across districts (Colombo, Kandy, Galle).
- Compare health improvements across age groups (e.g., ≤ 50 , 51–60, > 60).
- Examine differences in outcomes between males and females.

Explore Relationships with Baseline Risk Factors

- Investigate whether participants with diabetes, smoking history, or high baseline BP experienced different levels of improvement.
- Assess how baseline BMI, education level, and district relate to post-program health outcomes.
- Explore possible interactions between demographic characteristics (e.g., age $\times$ gender, district $\times$ education) and health improvements.

Descriptive and Summary Objectives

- Provide summary statistics (means, medians, ranges) for continuous variables such as age, blood pressure, and BMI.
- Generate counts and frequencies for categorical variables such as gender, district, education, smoking, diabetes, and participation.
- Summarize participation levels, including attendance at program sessions.

Association and Statistical Testing Objectives

- Identify associations between participation status and changes in risk factors (BP, BMI, smoking, exercise).
- Examine correlations between continuous risk factors (e.g., BMI and BP).
- Test whether program participation reduced the risk of CVD events after adjusting for baseline risk factors.

03. Data description

- ID: Unique code (CVD001–CVD1500)
- Age: Age of the participants(25-80 years)
- Gender: Gender of the participants
- District: The district the participant belonged to(Colombo, Kandy or Galle)
- Education: Education level of the participants in each district
- Pre_Systolic_BP: Systolic BP measured pre-program
- Post_Systolic_BP: Systolic BP measured post-program
- Pre_BMI: BMI measured pre-program
- Post_BMI: BMI measured post-program
- Smoking: If the participant is a smoker or not (Yes, No)
- Post_Smoking_Status: Smoking status of the participant post-program
- Pre_Exercise_Days: Number of exercise days per week pre-program
- Post_Exercise_Days: Number of exercise days per week post-program
- Diabetes: If the participant has diabetes or not (Yes or No)
- Participated: Status of participation
- Sessions_Attended: Number of sessions attended
- CVD_Event: Indicator if the participant experienced a CVD event during follow-up
- CVD_Prob:The probability of a participant experiencing a CVD based on risk factors

04. Data analysis and interpretation

This dataset contains 18 variables out of which the variables Age, Pre_Systolic_BP, Post_Systolic_BP, Pre_BMI, Post_BMI, Pre_Exercise_Days, Post_Exercise_Days, Sessions_Attended, CVD_Prob are continuous variables and the other variables ID, Gender, District, Education, Smoking, Post_Smoking_Status, Diabetes, Participated and CVD_Event are categorical variables. The below summary shows the summary for all continuous and categorical variables.

```
> summary(dataset)
```

ID	Age	Gender	District	Education
Length:1500	Min. :25.00	Length:1500	Length:1500	Length:1500
Class :character	1st Qu.:41.00	Class :character	Class :character	Class :character
Mode :character	Median :50.00	Mode :character	Mode :character	Mode :character
	Mean :50.13			
	3rd Qu.:59.00			
	Max. :80.00			
Pre_Systolic_BP	Post_Systolic_BP	Pre_BMI	Post_BMI	Smoking
Min. :110.0	Min. :100.0	Min. :18.50	Min. :18.50	Length:1500
1st Qu.:130.0	1st Qu.:121.0	1st Qu.:23.80	1st Qu.:22.50	Class :character
Median :142.0	Median :134.0	Median :26.70	Median :25.40	Mode :character
Mean :143.1	Mean :134.7	Mean :26.86	Mean :25.63	
3rd Qu.:155.2	3rd Qu.:147.0	3rd Qu.:29.70	3rd Qu.:28.40	
Max. :197.0	Max. :193.0	Max. :39.10	Max. :39.10	
Post_Smoking_Status	Pre_Exercise_Days	Post_Exercise_Days	Diabetes	
Length:1500	Min. :0.00	Min. :0.000	Length:1500	
Class :character	1st Qu.:1.00	1st Qu.:2.000	Class :character	
Mode :character	Median :2.00	Median :4.000	Mode :character	
	Mean :1.97	Mean :3.659		
	3rd Qu.:3.00	3rd Qu.:5.000		
	Max. :4.00	Max. :7.000		
Participated	Sessions_Attended	CVD_Event	CVD_Prob	
Length:1500	Min. :0.000	Length:1500	Min. :0.01000	
Class :character	1st Qu.:0.000	Class :character	1st Qu.:0.03000	
Mode :character	Median :7.000	Mode :character	Median :0.06500	
	Mean :5.333		Mean :0.06973	
	3rd Qu.:9.000		3rd Qu.:0.10000	
	Max. :12.000		Max. :0.23000	

We can observe the counts of categorical variable levels from creating the frequency tables by using the table function

Variable		Count	Proportion
Gender	Male	790	0.4733333
	Female	710	0.5266667
District	Colombo	587	0.3913333
	Galle	366	0.2440000
	Kandy	547	0.3646667
Education	Higher	321	0.2140000
	Primary	461	0.3073333
	Secondary	718	0.4786667
Smoking	No	1163	0.7753333
	Yes	337	0.2246667
Post_Smoking_Status	No	1233	0.822
	Yes	267	0.178
Diabetes	No	1232	0.8213333
	Yes	268	0.1786667
Participated	No	606	0.404
	Yes	894	0.596
CVD_Event	No	1405	0.9366667
	Yes	95	0.0633333

By the above proportions table we can observe that this dataset has more females than males that is there are 47.33% males and 52.67% females in the dataset. We can also observe that the highest proportion of people are from Colombo and an almost similar proportion slightly less than Colombo are from Kandy and the lowest proportion is from Galle. We can observe that the largest proportion has had secondary education and the proportion of people with higher education is the least. Out of the people participated a proportion of 0.2247 that is 22.47% have been smokers while the rest has been non smokers. We can observe that only 17.8% of the people smoke after the program while 82.2% of the people don't smoke after the program. This indicated that there has been an improvement of the proportion of non smokers after the program. It can be observed that 82.13% have not had diabetes while 17.87% has had diabetes. A proportion of 0.596 have participated for the program while 0.404 have not participated. 93.67% of the participants have not had any CVD event post follow up, while 0.053% has had a CVD event post follow up.

Further we can obtain frequency tables to summarize the continuous variables in our dataset.

- Age

```
> age_freq=ds_freq_table(dataset, Age, 5)
> age_freq
```

Variable: Age

Bins	Frequency	Cum Frequency	Percent	Cum Percent
25 - 36	221	221	14.73	14.73
36 - 47	447	668	29.8	44.53
47 - 58	515	1183	34.33	78.87
58 - 69	302	1485	20.13	99
69 - 80	133	1618	8.87	107.87
Total	1500	-	100.00	-

- Pre Systolic BP

```
> pre_bp_freq=ds_freq_table(dataset, Pre_Systolic_BP, 5)
> pre_bp_freq
```

Variable: Pre_Systolic_BP

Bins	Frequency	Cum Frequency	Percent	Cum Percent
110 - 127.4	324	324	21.6	21.6
127.4 - 144.8	492	816	32.8	54.4
144.8 - 162.2	461	1277	30.73	85.13
162.2 - 179.6	186	1463	12.4	97.53
179.6 - 197	37	1500	2.47	100
Total	1500	-	100.00	-

- Post Systolic BP

```
> post_bp_freq=ds_freq_table(dataset, Post_Systolic_BP, 5)
> post_bp_freq
```

Variable: Post_Systolic_BP

Bins	Frequency	Cum Frequency	Percent	Cum Percent
100 - 118.6	331	331	22.07	22.07
118.6 - 137.2	507	838	33.8	55.87
137.2 - 155.8	447	1285	29.8	85.67
155.8 - 174.4	180	1465	12	97.67
174.4 - 193	35	1500	2.33	100
Total	1500	-	100.00	-

- Pre BMI

```
> pre_bmi_freq=ds_freq_table(dataset,Pre_BMI,5)
> pre_bmi_freq
```

Variable: Pre_BMI

Bins	Frequency	Cum Frequency	Percent	Cum Percent
18.5 - 22.6	249	249	16.6	16.6
22.6 - 26.7	506	755	33.73	50.33
26.7 - 30.9	492	1247	32.8	83.13
30.9 - 35	205	1452	13.67	96.8
35 - 39.1	48	1500	3.2	100
Total	1500	-	100.00	-

- Post BMI

```
> post_bmi_freq=ds_freq_table(dataset,Post_BMI,5)
> post_bmi_freq
```

Variable: Post_BMI

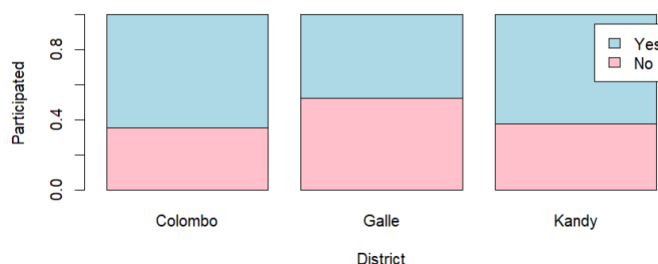
Bins	Frequency	Cum Frequency	Percent	Cum Percent
18.5 - 22.6	399	399	26.6	26.6
22.6 - 26.7	516	915	34.4	61
26.7 - 30.9	420	1335	28	89
30.9 - 35	140	1475	9.33	98.33
35 - 39.1	25	1500	1.67	100
Total	1500	-	100.00	-

05. Descriptive plots of qualitative data

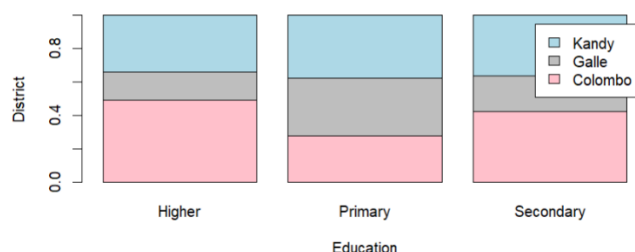
We can use descriptive plots to get a better idea about the variables in our dataset and to prove the revelations of the variables made by the tables graphically.

Pie chart of participation

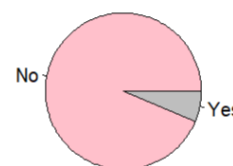
The proportion of people who had participated for the program are more than the people who did not participate for the program.

Compound barplot of Participated vs District

The number of people who participated are more than the people who didn't participate for Colombo and Kandy while the proportion of people who participated and did not participate are similar for Galle

Compound barplot of District vs Education

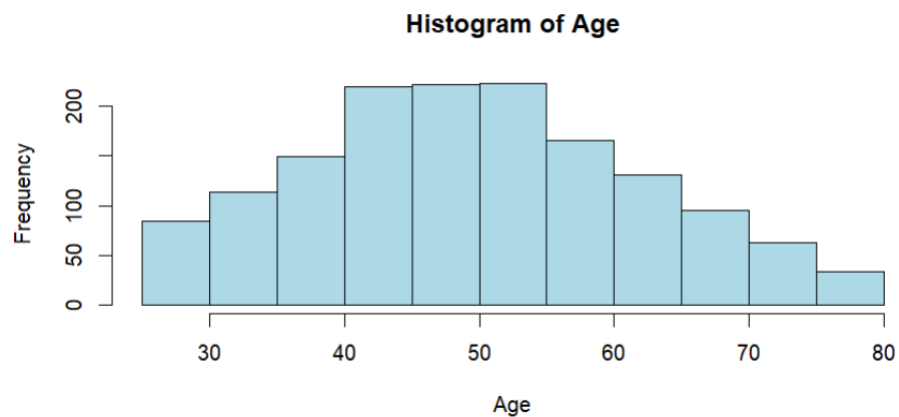
The most number of people who have had a higher education are from Colombo while the least are from Galle. Out of the people who have had a primary education the most number of people are from Kandy while the least are from Colombo. Out of the people who have had a secondary education the most are from Colombo while the least are from Galle.

Pie chart of having a CVD event

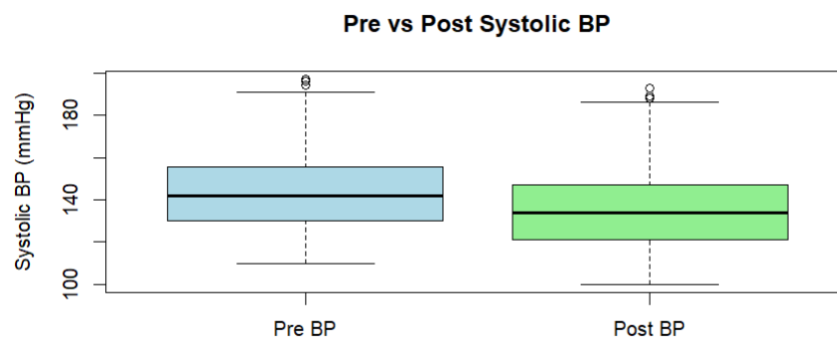
A large proportion of people have not had a CVD event, while only a really small proportion have had a CVD event

06. Descriptive plots of quantitative data

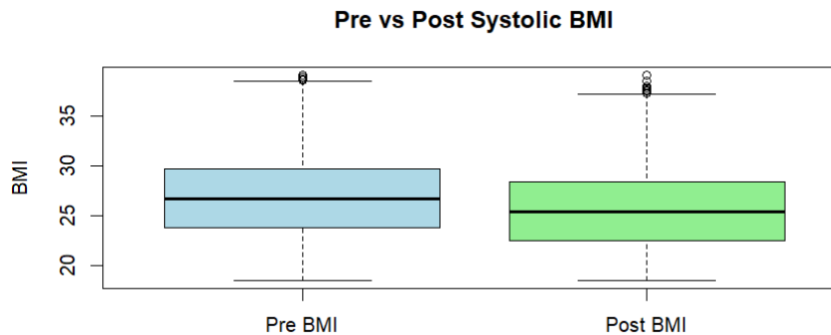
01) In order to study the distribution of the ages in the dataset we can plot a histogram.



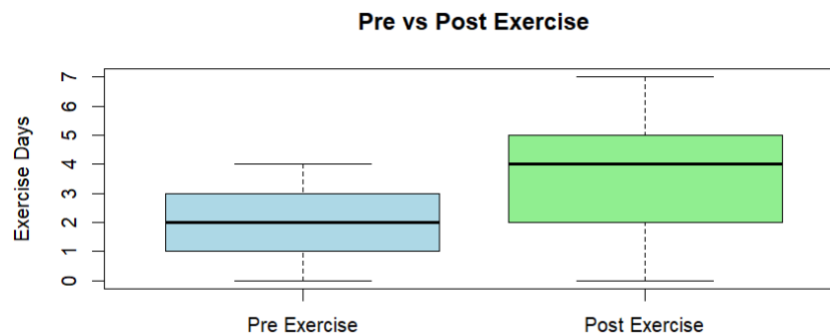
02) By using the below boxplot we can compare the pre systolic blood pressure and the post systolic blood pressure. It can be observed that mean and the interquartile range has decreased post program which is a healthy improvement. Some outliers can be observed.



- 03) The box plot between pre BMI and post BMI shows that the median value of post BMI has decreased slightly and the interquartile range has also shifted lower which indicates that people have started to reduce their weights according to their heights and this is an extremely healthy outcome. Some outliers can be observed here.



- 04) We can very clearly observe that the number of days of exercise post program has increased dramatically. The median has increased and the interquartile range has expanded which indicates that the number of people who have started exercising has also increased. No outliers are observed.



07. Normality Assumption

To check which variables could use parametric tests, we use the Shapiro test to get their normality accurately.

Shapiro Wilk Test

Ho: The data is normally distributed

H1: The data is not normally distributed

```
> #shapiro test
> shapiro.test(Pre_Systolic_BP)

        Shapiro-Wilk normality test

data:  Pre_Systolic_BP
W = 0.98453, p-value = 1.351e-11

> shapiro.test(Post_Systolic_BP)

        Shapiro-Wilk normality test

data:  Post_Systolic_BP
W = 0.98682, p-value = 1.924e-10

> shapiro.test(Pre_BMI)

        Shapiro-Wilk normality test

data:  Pre_BMI
W = 0.99066, p-value = 3.472e-08

> shapiro.test(Post_BMI)

        Shapiro-Wilk normality test

data:  Post_BMI
W = 0.98419, p-value = 9.226e-12
```

```
> shapiro.test(Pre_Exercise_Days)

        Shapiro-Wilk normality test

data:  Pre_Exercise_Days
W = 0.88829, p-value < 2.2e-16

> shapiro.test(Post_Exercise_Days)

        Shapiro-Wilk normality test

data:  Post_Exercise_Days
W = 0.95491, p-value < 2.2e-16
```

All the p-values are less than 0.05 we can reject Ho for all Variables.

Therefore we can conclude that all the variables do not follow a normal distribution. Since all the variables reject Ho, we should use Non-Parametric tests for the above variables instead of Parametric tests.

08. Impact of the program on health

To analyze the pre and post program changes in the systolic blood pressure, BMI and the number of exercise days, we use the non-parametric test, Wilcoxon's signed rank test.

- **Impact of the program on systolic blood pressure**

Ho : The median difference (Post_Systolic_BP – Pre_Systolic_BP) between the paired samples is zero.

H1 : The median difference (Post_Systolic_BP - Pre_Systolic_BP) between the paired samples is less than zero.

```
> wilcox.test(Post_Systolic_BP, Pre_Systolic_BP, mu=0, alternative="less", paired=T, exact=F, conf.in  
t=T, conf.level=0.95)
```

Wilcoxon signed rank test with continuity correction

```
data: Post_Systolic_BP and Pre_Systolic_BP  
V = 0, p-value < 2.2e-16  
alternative hypothesis: true location shift is less than 0  
95 percent confidence interval:  
-Inf -8.499953  
sample estimates:  
(pseudo)median  
-8.500045
```

P value($2.2e-16$) < 0.05, therefore we reject the null hypothesis at 5% level of significance.

General conclusion:

Therefore we can conclude that the median difference(Post_Systolic_BP-Pre_Systolic_BP) between the paired sample is less than zero. Therefore we can conclude that the Post_Systolic_BP is less than the Pre_Systolic_BP.

- **Impact of the program on BMI**

Ho : The median difference (Post_BMI – Pre_BMI) between the paired samples is zero.

H1 : The median difference (Post_BMI - Pre_BMI) between the paired samples is less than zero.

```
> wilcox.test(Post_BMI,Pre_BMI,mu=0,alternative="less",paired=T,exact=F,conf.int=T,conf.level=0.95)
```

Wilcoxon signed rank test with continuity correction

```
data: Post_BMI and Pre_BMI
V = 0, p-value < 2.2e-16
alternative hypothesis: true location shift is less than 0
95 percent confidence interval:
 -Inf -1.199982
sample estimates:
(pseudo)median
 -1.249981
```

P value($2.2e-16$) < 0.05, therefore we have enough evidence to reject H_0 at 5% level of significance.

General conclusion:

Therefore we can conclude that the median difference between post and pre BMI is less than zero. This concludes that the BMI has decreased post program.

- **Impact of the program on the number of exercise days**

H_0 : The median difference (Post_Exercise_Days – Pre_Exercise_Days) between the paired samples is zero.

H_1 : The median difference (Post_Exercise_Days - Pre_Exercise_Days) between the paired samples is greater than zero.

```
> wilcox.test(Post_Exercise_Days,Pre_Exercise_Days,mu=0,alternative="greater",paired=T,exact=F,conf.int=T,conf.level=0.95)
```

Wilcoxon signed rank test with continuity correction

```
data: Post_Exercise_Days and Pre_Exercise_Days
V = 707455, p-value < 2.2e-16
alternative hypothesis: true location shift is greater than 0
95 percent confidence interval:
 2.000016      Inf
sample estimates:
(pseudo)median
 2.000048
```

P value($2.2e-16$) < 0.05, therefore we have enough evidence to reject H_0 and 5% level of significance.

General conclusion:

Therefore we can conclude that the median difference between post and pre number of exercise days is greater than zero. This concludes that the number of exercise days have increased post program.

- **Impact of the program on health improvements across districts**

01. Systolic BP

Ho : There is no difference in the median Systolic BP improvement across the different districts

Ha : There is a difference in the median Systolic BP improvement in one or more districts.

```
> dataset$BP_Improvement <- dataset$Pre_Systolic_BP - dataset$Post_Systolic_BP  
> kruskal.test(dataset$BP_Improvement~District)
```

Kruskal-Wallis rank sum test

```
data: dataset$BP_Improvement by District  
Kruskal-Wallis chi-squared = 20.468, df = 2, p-value = 3.593e-05
```

The p value($3.593e-05$) <0.05 therefore we reject H0 at 5% level of significance.

General conclusion:

Therefore we have enough evidence to conclude that there is a difference in the Systolic BP improvement in one or more districts.

02. BMI

Ho : There is no difference in the median BMI improvement across the different districts

Ha : There is a difference in the median BMI improvement in one or more districts.


```
> dataset$BMI_Improvement <- dataset$Pre_BMI - dataset$Post_BMI  
> kruskal.test(dataset$BMI_Improvement~District)
```

Kruskal-Wallis rank sum test

```
data: dataset$BMI_Improvement by District  
Kruskal-Wallis chi-squared = 21.831, df = 2, p-value = 1.818e-05
```

The p value($1.818e-05$) <0.05 , therefore we reject H_0 at 5% level of significance.

General conclusion:

Therefore there is enough evidence to conclude that there is a difference in BMI improvement across the three districts.

03. Exercise days

H_0 : There is no difference in the median Exercise days improvement across the different districts

H_a : There is a difference in the median Exercise days improvement in one or more districts

```
> dataset$Exercise_Improvement <- dataset$Post_Exercise_Days - dataset$Pre_Exercise_Days  
> kruskal.test(dataset$Exercise_Improvement~District)
```

Kruskal-Wallis rank sum test

```
data: dataset$Exercise_Improvement by District  
Kruskal-Wallis chi-squared = 14.721, df = 2, p-value = 0.0006357
```

The p value(0.0006357) <0.05 , therefore we reject H_0 at 5% level of significance.

General conclusion:

Therefore we have enough evidence to conclude that there is a difference in the improvement of the number of days of exercise between the three districts.

- **Impact of the program on health improvements across ages**

In order to examine this the ages are first categorized into four groups as,

≤ 40 , 41–50, 51–60 and 61+

01. Systolic BP

H₀ : There is no difference in the median Systolic BP improvement across the different age groups

H_a : There is a difference in the median Systolic BP improvement in one or more age groups

```
> dataset$AgeGroup <- cut(dataset$Age,breaks = c(0, 40, 50, 60, 100),labels = c("<=40", "41-50",  
"51-60", "61+"),right = TRUE)  
> kruskal.test(dataset$BP_Improvement~dataset$AgeGroup)
```

Kruskal-Wallis rank sum test

data: dataset\$BP_Improvement by dataset\$AgeGroup

Kruskal-Wallis chi-squared = 1.3956, df = 3, p-value = 0.7066

The p value(0.7066)>0.05, therefore we do not reject H₀ at 5% level of significance.

General conclusion:

Therefore there is enough evidence to conclude that there is no difference in the systolic BP improvement across different age groups.

02. BMI

H₀ : There is no difference in the median BMI improvement across the different age groups

H_a : There is a difference in the median BMI improvement in one or more age groups

```
> kruskal.test(dataset$BMI_Improvement~dataset$AgeGroup)
```

Kruskal-Wallis rank sum test

data: dataset\$BMI_Improvement by dataset\$AgeGroup

Kruskal-Wallis chi-squared = 2.0464, df = 3, p-value = 0.5628

The p value(0.5628)>0.05, therefore we do not reject H₀ at 5% level of significance.

General conclusion:

Therefore there is enough evidence to conclude that there is no difference in the BMI improvement across different age groups.

03. Exercise days

H₀ : There is no difference in the median Exercise days improvement across the different age groups

H_a : There is a difference in the median Exercise days improvement in one or more age groups

```
> kruskal.test(dataset$Exercise_Improvement~dataset$AgeGroup)
```

```
Kruskal-Wallis rank sum test
```

```
data: dataset$Exercise_Improvement by dataset$AgeGroup
Kruskal-Wallis chi-squared = 3.6812, df = 3, p-value = 0.298
```

The p value(0.298)>0.05, therefore we do not reject H₀ at 5% level of significance.

General conclusion:

Therefore there is enough evidence to conclude that there is no difference in the exercise days improvement across different age groups.

- **Impact of the program on health improvements across gender**

01. Systolic BP

H₀ : There is no difference in the location of Systolic BP improvement across the gender

H_a : There is a difference in the location of Systolic BP improvement across the gender

```
> wilcox.test(dataset$BP_Improvement~Gender, mu = 0, alternative = "two.sided",conf.int = T,Conf.level =0.95,exact = F,correct = T)
```

```
Wilcoxon rank sum test with continuity correction
```

```
data: dataset$BP_Improvement by Gender
W = 275981, p-value = 0.593
alternative hypothesis: true location shift is not equal to 0
95 percent confidence interval:
 -9.999232e-01  2.999787e-05
sample estimates:
difference in location
 -2.28027e-05
```

The p value(0.593)>0.05, therefore we do not reject H0 at 5% level of significance.

General conclusion:

Therefore there is enough evidence to conclude that there is no difference in location in the systolic BP improvement across the gender.

02. BMI

H0 : There is no difference in the location of BMI improvement across the gender

Ha : There is a difference in the location of BMI improvement across the gender

```
> wilcox.test(dataset$BMI_Improvement~Gender, mu = 0, alternative = "two.sided",conf.int = T,Conf.level =0.95,exact = F,correct = T)
```

Wilcoxon rank sum test with continuity correction

```
data: dataset$BMI_Improvement by Gender
W = 267646, p-value = 0.1262
alternative hypothesis: true location shift is not equal to 0
95 percent confidence interval:
 -1.000268e-01  7.210285e-05
sample estimates:
difference in location
 -0.09993191
```

The p value(0.1262)>0.05, therefore we do not reject H0 at 5% level of significance.

General conclusion:

Therefore there is enough evidence to conclude that there is no difference in location in the BMI improvement across the gender.

03. Exercise days

H0 : There is no difference in the location of Exercise days improvement across the gender

Ha : There is a difference in the location of Exercise days improvement across the gender

```
> wilcox.test(dataset$Exercise_Improvement~Gender, mu = 0, alternative = "two.sided",conf.int = T,Conf.level =0.95,exact = F,correct = T)
```

Wilcoxon rank sum test with continuity correction

```
data: dataset$Exercise_Improvement by Gender
W = 278922, p-value = 0.8509
alternative hypothesis: true location shift is not equal to 0
95 percent confidence interval:
 -2.183171e-05  5.783596e-05
sample estimates:
difference in location
 -3.864237e-05
```

The p value(0.8509)>0.05, therefore we do not reject H0 at 5% level of significance.

General conclusion:

Therefore there is enough evidence to conclude that there is no difference in location in the exercise days improvement across the gender.

- **Checking correlations**

01. Between diabetes and CVD event

H0: Diabetes and CVD event are independent

Ha: Diabetes and CVD event are dependent

```
> table5=table(Diabetes,CVD_Event)
> chisq.test(table5,correct=T)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: table5
X-squared = 3.2622, df = 1, p-value = 0.07089
```

P value(0.07089)>0.05, therefore we do not reject H0 at 5% level of significance.

General conclusion:

Therefore we have enough evidence to conclude that Diabetes and CVD event are independent.

02. Between post smoking status and CVD event

H0: Post smoking status and CVD event are independent

Ha: Post smoking status and CVD event are dependent

```
> table8=table(Post_Smoking_Status,CVD_Event)
> chisq.test(table8,correct=T)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: table8
X-squared = 12.174, df = 1, p-value = 0.0004845
```

P value(0.00048)<0.05, therefore we reject H0 at 5% level of significance.

General conclusion:

Therefore we have enough evidence to conclude that Post smoking status and CVD event are not independent.

03. Between participation and CVD event

H0: Participation and CVD event are independent

Ha: Participation and CVD event are dependent

```
> table7=table(Participated,CVD_Event)
> chisq.test(table7,correct = T)
```

Pearson's Chi-squared test with Yates' continuity correction

```
data: table7
X-squared = 1.2235, df = 1, p-value = 0.2687
```

P value(0.2687)>0.05, therefore we do not reject H0 at 5% level of significance.

General conclusion:

Therefore we have enough evidence to conclude that participation and CVD event are independent.

09. Conclusion

This community-based CVD prevention program demonstrated measurable improvements in several key cardiovascular risk factors among participants across Colombo, Kandy, and Galle. Reductions in systolic blood pressure and body mass index, increased exercise frequency, and notable rates of smoking reduction indicates that structured lifestyle behaviors can effectively promote healthier lifestyles in the community.

The Kruskal Wallis tests revealed the presence of some variation in the improvements of systolic blood pressure, BMI and number of exercise days between districts, suggesting that local factors of districts may influence program outcomes. We were also able to examine that the improvements in health factors were neither affected by the various age groups nor by the gender. Correlation and association analyses also highlighted meaningful relationships between baseline risk factors such as diabetes, post smoking status and the participation for the event with if a CVD event had occurred in the follow-up. We were able to find out that while the post smoking status was correlated with the CVD event, participation and diabetes were independent of the CVD event happening.

Overall, the results show that community-based programs with multiple components can make a real difference in helping people manage their heart health. By focusing on lifestyle changes and regular follow-ups, these programs can lower common risk factors and reduce the chances of future heart disease. Expanding and adapting similar initiatives across more districts could play an important role in improving heart health and easing the growing burden of cardiovascular disease in Sri Lanka.

10. R code

```
#creating the dataset
```

```
set.seed(16661)
```

```
if (!requireNamespace("truncnorm", quietly = TRUE)) {
```

```
  install.packages("truncnorm")
```

```
}
```

```
library(truncnorm)
```

```
n <- 1500
```

```
ID <- sprintf("CVD%03d", 1:n)
```

```
Age <- round(rtruncnorm(n, a = 25, b = 80, mean = 50, sd = 15))
```

```
Gender <- sample(c("Female", "Male"), size = n, replace = TRUE, prob = c(0.52, 0.48))
```

```
District <- sample(c("Colombo", "Kandy", "Galle"), size = n, replace = TRUE, prob = c(0.40, 0.35, 0.25))
```

```
Education <- character(n)
```

```
for (d in c("Colombo", "Kandy", "Galle")) {
```

```
  idx <- which(District == d)
```

```
  if (d == "Colombo") {
```

```
    Education[idx] <- sample(c("Primary", "Secondary", "Higher"), length(idx), replace = TRUE, prob = c(0.20, 0.50, 0.30))
```

```
  } else if (d == "Kandy") {
```

```
    Education[idx] <- sample(c("Primary", "Secondary", "Higher"), length(idx), replace = TRUE, prob = c(0.35, 0.45, 0.20))
```

```
  } else if (d == "Galle") {
```

```
    Education[idx] <- sample(c("Primary", "Secondary", "Higher"), length(idx), replace = TRUE, prob = c(0.45, 0.40, 0.15))
```

```
  }
```

```
}
```

```
Pre_Systolic_BP <- round(rtruncnorm(n, a = 110, b = 200, mean = 140, sd = 20))
```

```
Pre_BMI <- round(rtruncnorm(n, a = 18.5, b = 40, mean = 26.5, sd = 4.5), 1)
```

```
Smoking <- character(n)
```

```
for (i in seq_len(n)) {
```

```
  if (Gender[i] == "Male") {
```

```
    Smoking[i] <- ifelse(runif(1) < 0.35, "Yes", "No")
```

```
  } else {
```

```
    Smoking[i] <- ifelse(runif(1) < 0.08, "Yes", "No")
```

```
  }
```



```
}  
  
Pre_Exercise_Days <- sample(0:4, size = n, replace = TRUE)  
  
Diabetes <- ifelse(Age > 50,  
  ifelse(runif(n) < 0.25, "Yes", "No"),  
  ifelse(runif(n) < 0.12, "Yes", "No"))  
  
Participation_prob <- ifelse(District == "Colombo", 0.70,  
  ifelse(District == "Kandy", 0.60, 0.50))  
  
Participated <- ifelse(runif(n) < Participation_prob, "Yes", "No")  
  
Sessions_Attended <- ifelse(Participated == "Yes", sample(6:12, n, replace = TRUE), 0)  
  
improvement_bp <- numeric(n)  
  
for (i in seq_len(n)) {  
  if (Participated[i] == "Yes") {  
    improvement_bp[i] <- runif(1, min = 5, max = 20)  
  } else {  
    improvement_bp[i] <- runif(1, min = 0, max = 5)  
  }  
}  
  
Post_Systolic_BP <- pmax(100, round(Pre_Systolic_BP - improvement_bp))  
  
improvement_bmi <- numeric(n)  
  
for (i in seq_len(n)) {  
  if (Participated[i] == "Yes") {  
    improvement_bmi[i] <- runif(1, min = 0.5, max = 3.0)  
  } else {  
    improvement_bmi[i] <- runif(1, min = 0.0, max = 1.0)  
  }  
}  
  
Post_BMI <- round(pmax(18.5, Pre_BMI - improvement_bmi), 1)  
  
Post_Smoking_Status <- Smoking  
  
for (i in seq_len(n)) {  
  if (Smoking[i] == "Yes") {  
    if (Participated[i] == "Yes") {  
      Post_Smoking_Status[i] <- ifelse(runif(1) < 0.30, "No", "Yes")  
    } else {  
      Post_Smoking_Status[i] <- "Yes"  
    }  
  } else {  
    Post_Smoking_Status[i] <- "No"  
  }  
}
```

```
Post_Smoking_Status[i] <- ifelse(runif(1) < 0.10, "No", "Yes")
}
} else {
  Post_Smoking_Status[i] <- "No"
}
}

Post_Exercise_Days <- integer(n)
for (i in seq_len(n)) {
  if (Participated[i] == "Yes") {
    inc <- sample(1:4, 1)
  } else {
    inc <- sample(0:1, 1)
  }
  Post_Exercise_Days[i] <- pmin(7, Pre_Exercise_Days[i] + inc)
}

base_prob <- 0.03
cvd_prob <- rep(base_prob, n)
cvd_prob <- cvd_prob + ifelse(Age > 60, 0.04, 0)
cvd_prob <- cvd_prob + ifelse(Gender == "Male", 0.03, 0)
cvd_prob <- cvd_prob + ifelse(Smoking == "Yes", 0.05, 0)    # using pre-program smoking
cvd_prob <- cvd_prob + ifelse(Diabetes == "Yes", 0.06, 0)
cvd_prob <- cvd_prob + ifelse(Pre_Systolic_BP > 160, 0.04, 0)
cvd_prob <- cvd_prob - ifelse(Participated == "Yes", 0.02, 0)
cvd_prob <- pmin(pmax(cvd_prob, 0), 1)
CVD_Event <- ifelse(runif(n) < cvd_prob, "Yes", "No")

dataset <-
data.frame(ID, Age, Gender, District, Education, Pre_Systolic_BP, Post_Systolic_BP, Pre_BMI, Post_BMI, Smoking, Post_Smoking_Status, Pre_Exercise_Days, Post_Exercise_Days, Diabetes, Participated, Sessions_Attended, CVD_Event, CVD_Prob = round(cvd_prob, 4), stringsAsFactors = FALSE)

cat("Number of rows:", nrow(dataset), "\n")
cat("Participation counts:\n"); print(table(dataset$Participated))
cat("Smoking (pre) counts:\n"); print(table(dataset$Smoking))
cat("CVD event counts:\n"); print(table(dataset$CVD_Event))

write.csv(dataset, "synthetic_cvd_dataset_1500.csv", row.names = FALSE)
```

```
cat("Saved dataset to 'synthetic_cvd_dataset_1500.csv' in working directory.\n")
```

```
head(dataset, 8)
```

```
#summary of the dataset
```

```
summary(dataset)
```

```
#finding the counts
```

```
count_t_gender=table(Gender)
```

```
count_t_gender
```

```
count_t_district=table(District)
```

```
count_t_district
```

```
count_t_education=table(Education)
```

```
count_t_education
```

```
count_t_smoking=table(Smoking)
```

```
count_t_smoking
```

```
count_t_postsmoking=table(Post_Smoking_Status)
```

```
count_t_postsmoking
```

```
count_t_diabetes=table(Diabetes)
```

```
count_t_diabetes
```

```
count_t_participated=table(Participated)
```

```
count_t_participated
```

```
count_t_cvdevent=table(CVD_Event)
```

```
count_t_cvdevent
```

```
#proportions tables
```

```
prop_t_gender=table(Gender)/length(Gender)
```

```
prop_t_gender
```

```
prop_t_district=table(District)/length(District)
```

```
prop_t_district
```

```
prop_t_education=table(Education)/length(Education)
```

```
prop_t_education
```

```
prop_t_smoking=table(Smoking)/length(Smoking)
```

```
prop_t_smoking
```

```
prop_t_postsmoking=table(Post_Smoking_Status)/length(Post_Smoking_Status)
```

```
prop_t_postsmoking

prop_t_diabetes=table(Diabetes)/length(Diabetes)

prop_t_diabetes

prop_t_participated=table(Participated)/length(Participated)

prop_t_participated

prop_t_cvdevent=table(CVD_Event)/length(CVD_Event)

prop_t_cvdevent

#frequency tables

library(descriptr)

age_freq=ds_freq_table(dataset,Age,5)

age_freq

pre_bp_freq=ds_freq_table(dataset,Pre_Systolic_BP,5)

pre_bp_freq

post_bp_freq=ds_freq_table(dataset,Post_Systolic_BP,5)

post_bp_freq

pre_bmi_freq=ds_freq_table(dataset,Pre_BMI,5)

pre_bmi_freq

post_bmi_freq=ds_freq_table(dataset,Post_BMI,5)

post_bmi_freq

pre_ex_freq=ds_freq_table(dataset,Pre_Exercise_Days,5)

pre_ex_freq

post_ex_freq=ds_freq_table(dataset,Post_Exercise_Days,5)

post_ex_freq

session_freq=ds_freq_table(dataset,Sessions_Attended,5)

session_freq

#descriptive plots

pie(count_t_gender,main="Pie chart of Gender",col=c("pink","grey"))

barplot(count_t_district,main="Bar chart of district",xlab="District",ylab="Frequency",col=c("pink","grey","lightblue"))

barplot(count_t_education,main="Pie chart of education",xlab="Education",ylab="Frequency",col=c("pink","grey","lightblue"))

pie(count_t_smoking,main="Pie chart of smoking",col=c("pink","grey"))

pie(count_t_postsmoking,main="Pie chart of post smoking",col=c("pink","grey"))

pie(count_t_diabetes,main="Pie chart of diabetes",col=c("pink","grey"))

pie(count_t_participated,main="Pie chart of participation",col=c("pink","grey"))
```

```
pie(count_t_cvdevent,main="Pie chart of having a CVD event",col=c("pink","grey"))

Table1=table(District,Education)

table2=prop.table(Table1,2)

table2

barplot(table2,legend.text=row.names(Table1),main="Compound barplot of District vs
Education",xlab="Education",ylab="District",col=c("pink","grey","lightblue"))

table3=table(Participated, District)

table4=prop.table(table3,2)

table4

barplot(table4,legend.text=row.names(table3),main="Compound barplot of Participated vs
District",xlab="District",ylab="Participated",col=c("pink","lightblue"))

hist(Age,col="lightblue")

boxplot(dataset$Pre_Systolic_BP, dataset$Post_Systolic_BP,names = c("Pre BP", "Post BP"), col = c("lightblue", "lightgreen"),main
= "Pre vs Post Systolic BP",

      ylab = "Systolic BP (mmHg)")

boxplot(dataset$Pre_BMI, dataset$Post_BMI,names = c("Pre BMI", "Post BMI"), col = c("lightblue", "lightgreen"),main = "Pre vs
Post Systolic BMI",

      ylab = "BMI")

boxplot(dataset$Pre_Exercise_Days, dataset$Post_Exercise_Days,names = c("Pre Exercise", "Post Exercise"), col = c("lightblue",
"lightgreen"),main = "Pre vs Post Exercise",

      ylab = "Exercise Days")

#shapiro test

shapiro.test(Pre_Systolic_BP)

shapiro.test(Post_Systolic_BP)

shapiro.test(Pre_BMI)

shapiro.test(Post_BMI)

shapiro.test(Pre_Exercise_Days)

shapiro.test(Post_Exercise_Days)

#nonparametric tests

wilcox.test(Post_Systolic_BP,Pre_Systolic_BP,mu=0,alternative="less",paired=T,exact=F,conf.int=T,conf.level=0.95)

wilcox.test(Post_BMI,Pre_BMI,mu=0,alternative="less",paired=T,exact=F,conf.int=T,conf.level=0.95)

wilcox.test(Post_Exercise_Days,Pre_Exercise_Days,mu=0,alternative="greater",paired=T,exact=F,conf.int=T,conf.level=0.95)

dataset$BP_Improvement <- dataset$Pre_Systolic_BP - dataset$Post_Systolic_BP

kruskal.test(dataset$BP_Improvement~District)
```

```
dataset$BMI_Improvement <- dataset$Pre_BMI - dataset$Post_BMI

kruskal.test(dataset$BMI_Improvement~District)

dataset$Exercise_Improvement <- dataset$Post_Exercise_Days - dataset$Pre_Exercise_Days
kruskal.test(dataset$Exercise_Improvement~District)

dataset$AgeGroup <- cut(dataset$Age,breaks = c(0, 40, 50, 60, 100),labels = c("≤40", "41–50", "51–60", "61+"),right = TRUE)

kruskal.test(dataset$BP_Improvement~dataset$AgeGroup)

kruskal.test(dataset$BMI_Improvement~dataset$AgeGroup)

kruskal.test(dataset$Exercise_Improvement~dataset$AgeGroup)

wilcox.test(dataset$BP_Improvement~Gender, mu = 0, alternative = "two.sided",conf.int = T,Conf.level =0.95,exact = F,correct = T)

wilcox.test(dataset$BMI_Improvement~Gender, mu = 0, alternative = "two.sided",conf.int = T,Conf.level =0.95,exact = F,correct = T)

wilcox.test(dataset$Exercise_Improvement~Gender, mu = 0, alternative = "two.sided",conf.int = T,Conf.level =0.95,exact = F,correct
= T)

#checking correlations

table5=table(Diabetes,CVD_Event)

chisq.test(table5,correct=T)

table7=table(Participated,CVD_Event)

chisq.test(table7,correct = T)

table8=table(Post_Smoking_Status,CVD_Event)

chisq.test(table8,correct=T)
```

The dataset created:

<https://docs.google.com/spreadsheets/d/1y2yZgzqu9-rRyOk8bpnavslu-JMbD8Fs/edit?usp=sharing&oid=105960429400668151794&rtpof=true&sd=true>